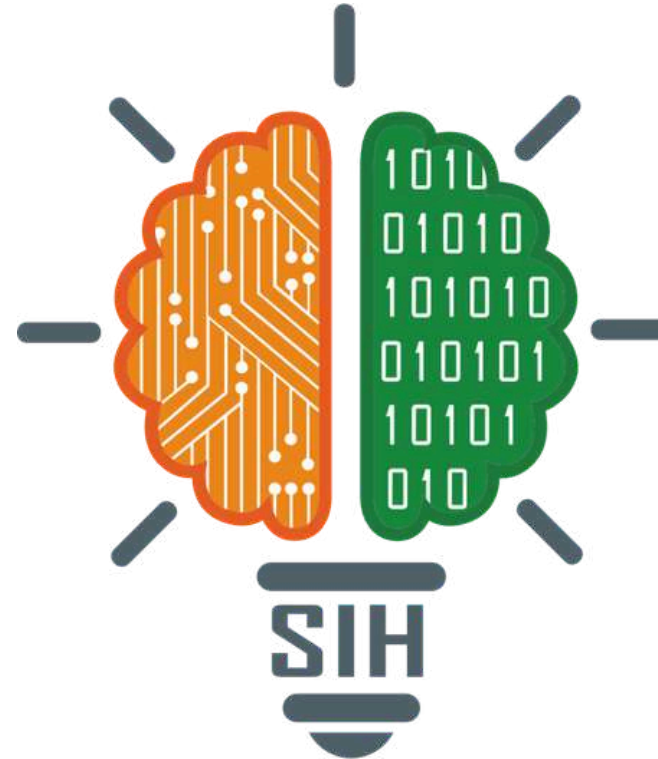


SMART INDIA HACKATHON 2026



SMART INDIA
HACKATHON
2026

- Problem Statement ID – SIH26182
- Problem Statement Title – Automated Attribution of Unknown Cryptocurrency Wallets to Nearest VASPs through Blockchain Intelligence APIs
- Theme – Blockchain & Cybersecurity
- PS Category – Software
- Team ID –
- Team Name – TraceX



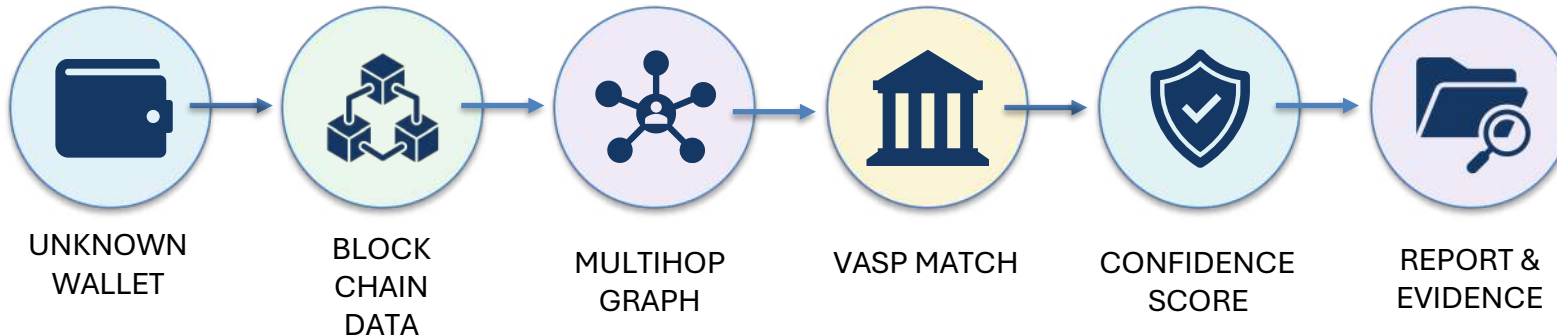
Proposed Solution

PROBLEM

- Unknown crypto wallets may appear during cybercrime investigations.
- Manually tracing transactions path can be complex and time consuming.
- Identifying the nearest VASP requires blockchain analysis.

OUR SOLUTION

- Trace: Follow the wallet's blockchain transactions.
- Match: Find connections with known VASP addresses.
- Score: Generate an explainable confidence score.
- Report: Provide transaction path and investigation-ready evidence.



INNOVATION

- Multi-chainwallet graph tracing
- Labelled VASP address matching.
- Explainable confidence scoring
- SAHYOG-workflow-ready disclosure draft.

TECHNOLOGIES TO BE USED

BACKEND

- Python + FastAPI

FRONTEND

- React

DATABASE

- PostgreSQL

**BLOCKCHAIN DATA
APIs**

- Bitquery
- Etherscan
- Transcan

**GRAPH
TRAVERSAL**

- BFS/DFS

REPORTING

- JSON + PDF

WORKING METHODOLOGY & PROCESS FOR IMPLEMENTATION

INPUT


**SUSPECT WALLET +
CHAIN**
Wallet address +
Blockchain

PROCESS

ATTRIBUTION ENGINE

BLOCKCHAIN DATA API

MULTI-HOP GRAPH TRAVERSAL

LABELLED VASP ADDRESS MATCHING

CONFIDENCE SCORE + RISK FLAGGING

OUTPUT

**INVESTIGATOR
DASHBOARD**
Path visualization +
Review

JSON/PDF REPORT
SAHYOG
DISCLOSURE DRAFT

FEASIBILITY

- ✓ API-bases architecture
- ✓ Ethereum + Tron MVP
- ✓ Multi-hop graph traversal
- ✓ Existing blockchain data APIs
- ✓ Human-in-the-loop review

POTENTIALCHALLENGES & RISKS

- Sparse labelled dataset
- Uncalibrated confidence score
- Blockchain API rate limits
- SAHYOG integration delays
- Transaction obfuscation
- Risk of over-trust

STRATEGIES FOR OVERCOMING

- Curated labelled addresses
- Clearly label score as heuristic
- API rate-limit handling
- Mock SAHYOG payload during MVP
- Flag mixers/bridges/high fan-out
- Mandatory investigator review

GO/ NO-GO VALIDATION

Trace Accuracy

Attribution Usefulness

Investigator Usability

TARGET USERS/IMPACT

State/UT Cyber cells / I4C / ED/CBI / VASP Nodal officers



FASTER ATTRIBUTION

Reduce manual tracing effort by automating multi-hop wallet tracing and VASP matching.



INVESTIGATION SUPPORT

Provide explainable transaction paths with confidence scoring and risk flags for investigator review.



STANDARDIZED EVIDENCE

Generate structured JSON/PDF reports with transaction paths and investigate ready evidence.

WORKFLOW ENABLEMENT

SAHYOG-ready disclosure request

Human-in-the-loop : Investigate lead, not legal proof

BLOCKCHAIN & CRYPTOCRIME

S1 — Chainalysis
2026 Crypto Crime
Report
S2 — TRM Labs
Forensics & AI-
assisted investigation
tooling

BLOCKCHAIN ANALYTICS

S3 — Elliptic
2026 Regulatory
Outlook/Lens
S4 — GraphSense
(IKNAIO)
Open-source
cryptocurrency
analytics platform

DATA & INFRASTRUCTURE

S5 — Blockchain Data
APIs

- Bitquery
- Etherscan
- TronScan
- BscScan
- SoLscan

INDIA / COMPLIANCE

S6 — MHA/14C
SAHYOG portal &
workflow context
S7 — Enforcement
Directorate
Public reporting on
Crypto-linked fraud
S8 — OFAC /
Opensanctions
Public sanctioned
address lists